STATIC TIMING ANALYSIS OF VSDBabySoC

SKY130 Multi-Corner OpenSTA — Complete Working & Evidence Report

Name	B.Rohith Reddy
Roll Number	25EG504B06
Department / Section	ECE / B
Design	VSDBabySoC
STA Tool	OpenSTA 3.1.0
Clock	clk — 11.00 ns
Corners Run	18
Environment	Ubuntu VM / VirtualBox + Docker

This report keeps the actual terminal evidence, intermediate debugging, successful STA runs, timing paths, multi-corner output, and raw copied terminal logs. The purpose is to document the work as it was performed in my own VM.

Prepared from my VSDBabySoC working directory and uploaded terminal evidence.

1. What I Did

I performed static timing analysis on the synthesized VSDBabySoC design using OpenSTA 3.1.0. I first checked the available files and libraries, debugged the initial linking problem, loaded the SKY130 standard-cell timing library, linked the synthesized netlist, applied the 11 ns clock constraint, and then ran setup and hold checks across the available 18 timing-library cases.

The evidence section intentionally includes the terminal screenshots with the Ubuntu/VirtualBox desktop visible, the commands I entered, warnings that appeared during the work, and the final completion message. The raw-output appendix contains the copied terminal logs rather than replacing them with only a cleaned-up summary.

2. Files Used

File	Purpose
src/lib/vsdbabysoc.synth.v	Synthesized VSDBabySoC netlist
src/lib/avsdpll.lib	PLL timing library
src/lib/avsddac.lib	DAC timing library
src/lib/sky130_fd_sc_hd__*.lib	SKY130 HD standard-cell timing libraries
src/sdc/vsdbabysoc_synthesis.sdc	Timing constraints / clock definition
18_corner_results.txt	Recorded multi-corner OpenSTA results

3. STA Setup Used

The clock constraint used for the working analysis was:

```
set_units -time ns
create_clock -name clk -period 11 [get_pins {pll/CLK}]
```

The normal OpenSTA sequence used for each corner was:

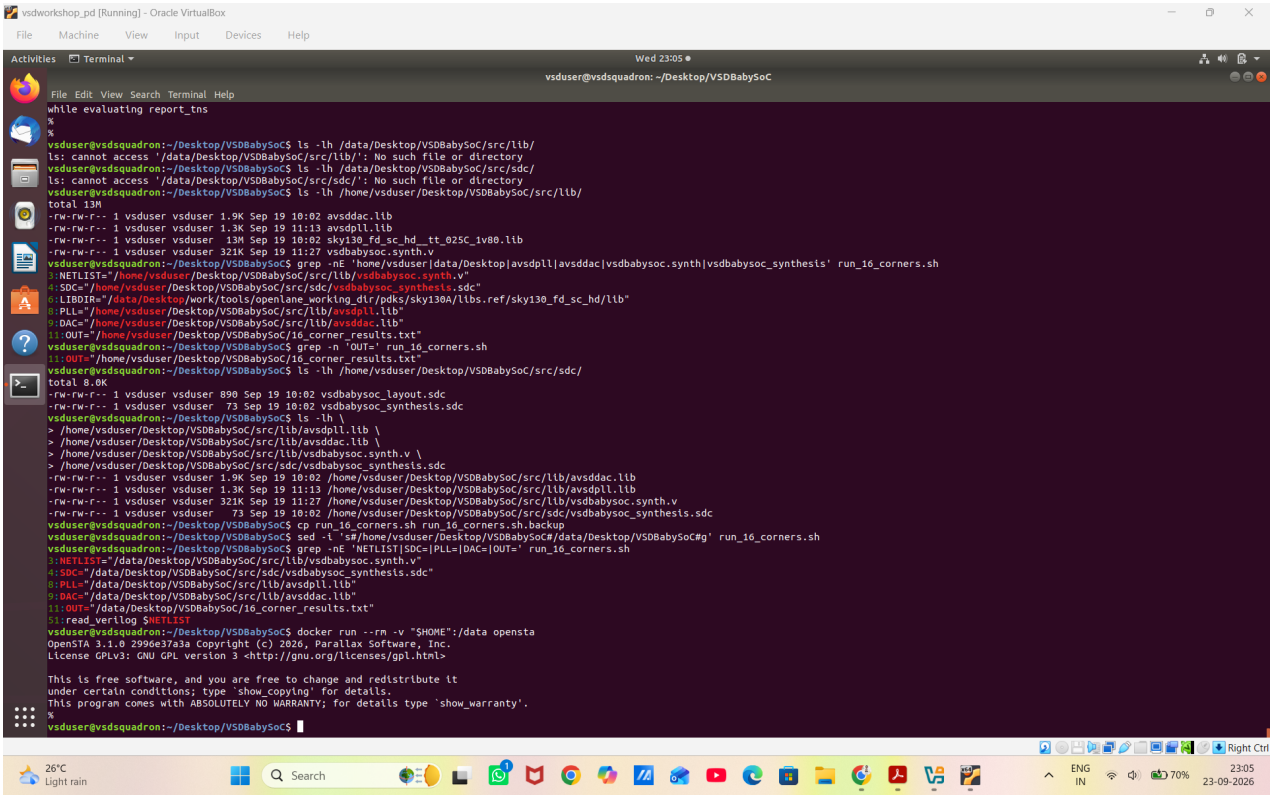
```
read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
read_liberty <corner SKY130 HD .lib>
read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
link_design vsdbabysoc
read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
report_checks -path_delay max -group_path_count 1 -format full_clock_expanded
report_checks -path_delay min -group_path_count 1 -format full_clock_expanded
report_wns
report_tns
report_clock_skew
```

During early runs, the standard-cell library was missing and OpenSTA created black boxes for many SKY130 cells. After loading the appropriate SKY130 HD Liberty file, the design linked and timing paths became available. The recurring default_fanout_load warning was retained in the raw evidence because it is part of the actual terminal output.

4. My Terminal Evidence

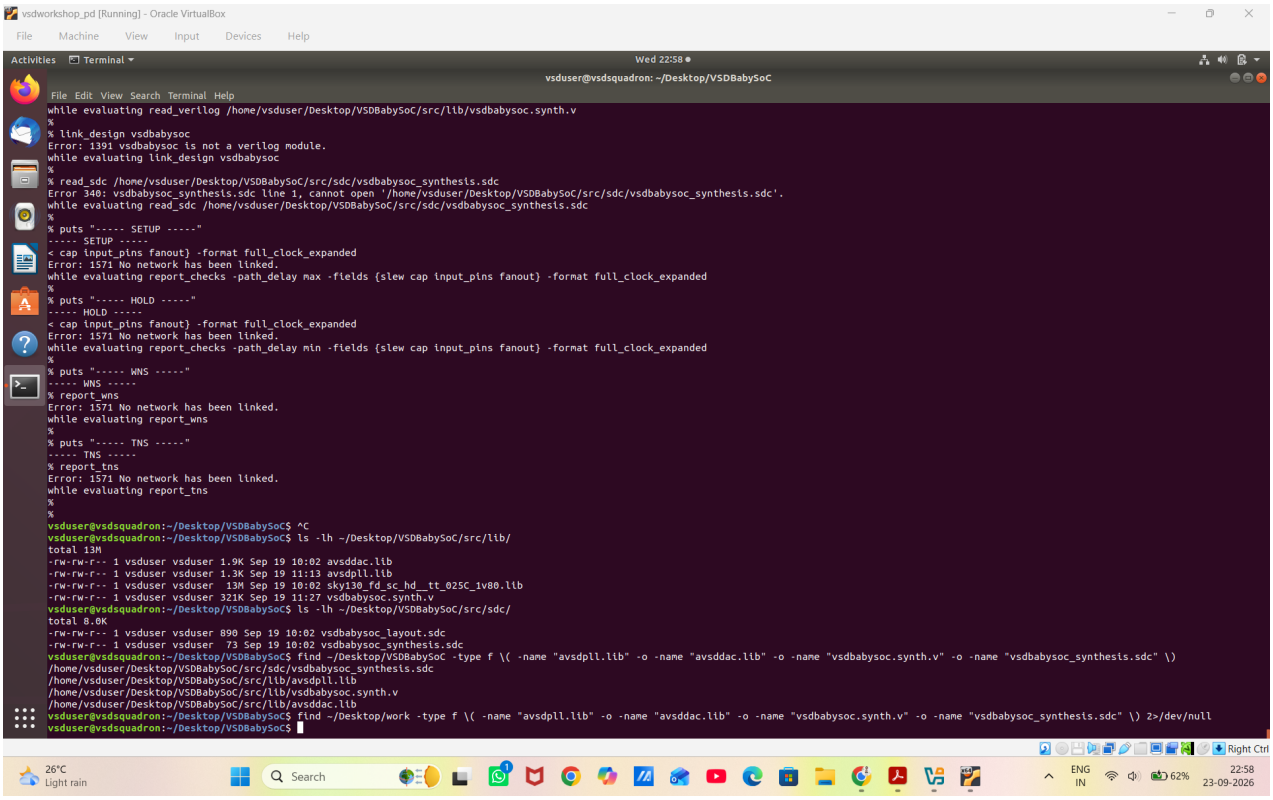
These screenshots are kept as evidence from the working session. They show the actual Ubuntu VM terminal, project path, commands, warnings, timing paths and final completion message.

Evidence Screenshot 1



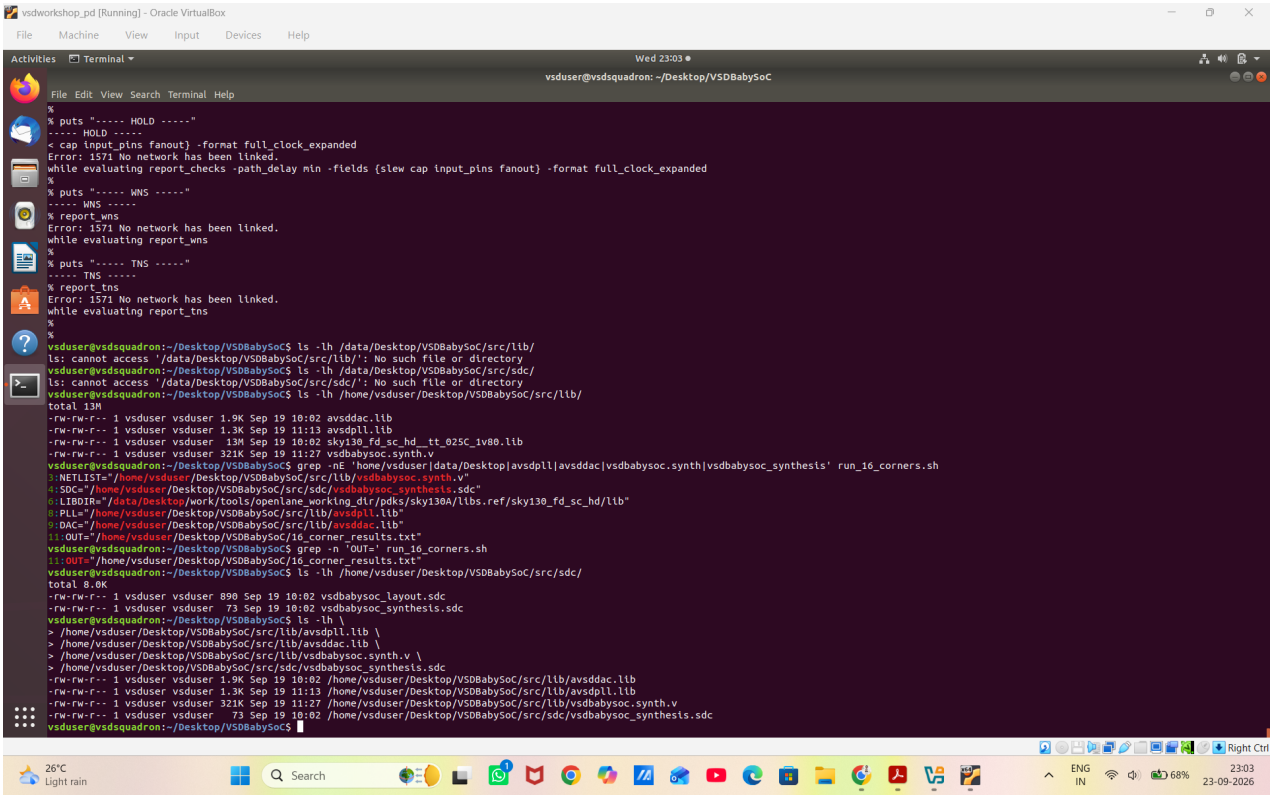
Initial working-directory and file/library checks, including the corner script and OpenSTA launch.

Evidence Screenshot 2



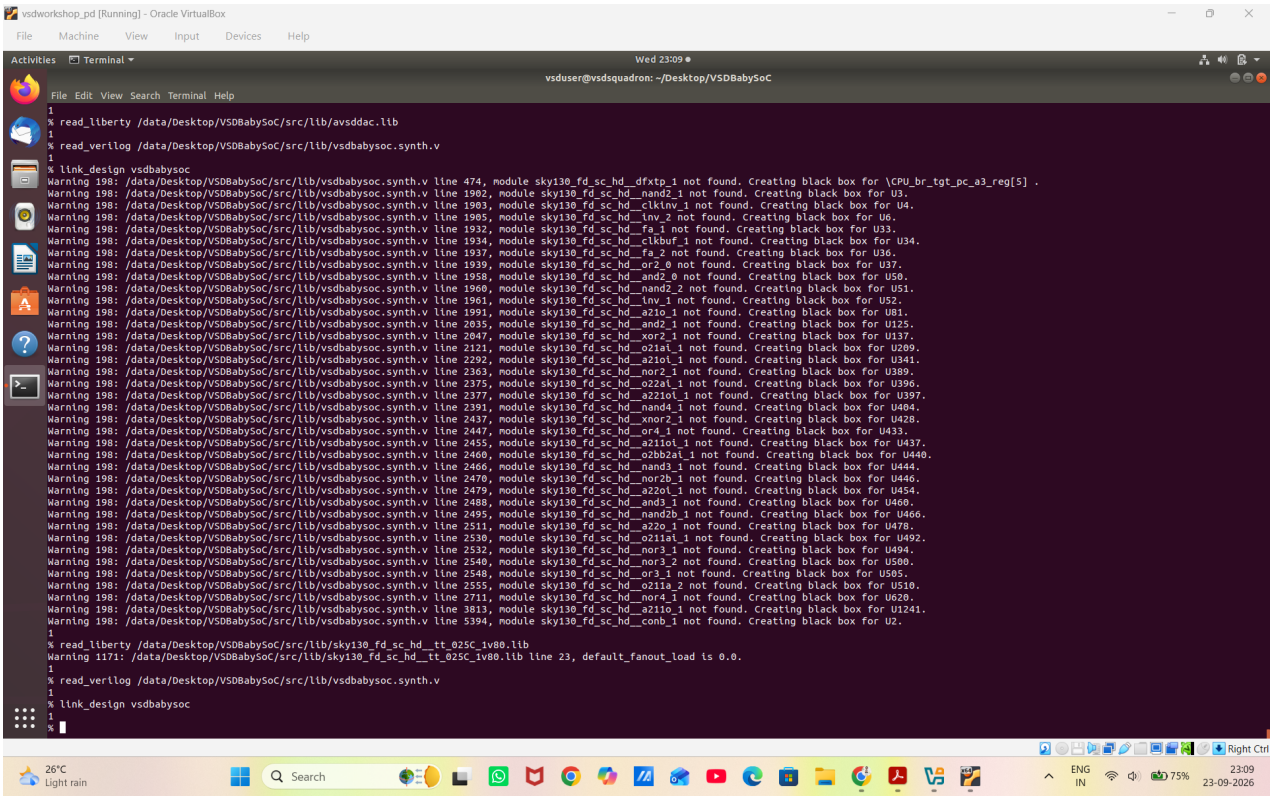
Early debugging: unresolved netlist/SDC path and network-not-linked messages.

Evidence Screenshot 3



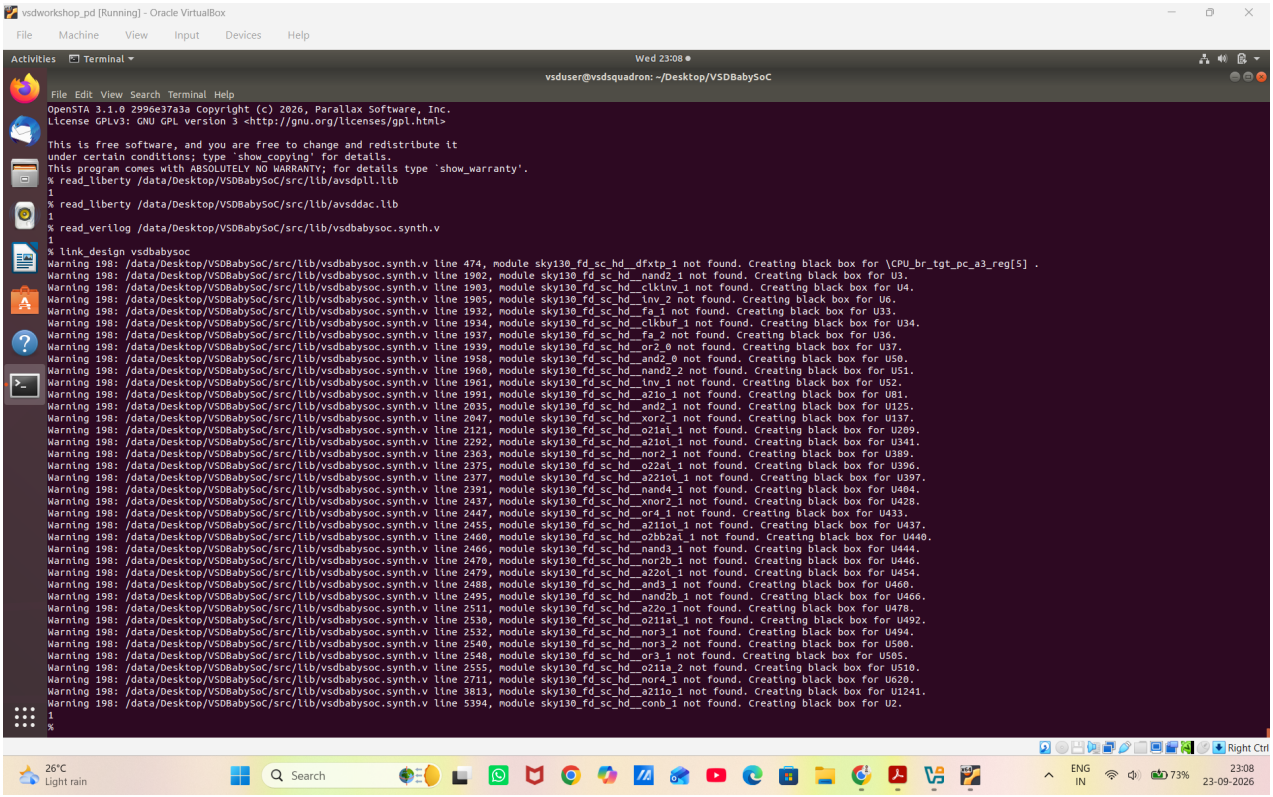
Early debugging after the first timing-report attempt, followed by checking the actual project files.

Evidence Screenshot 4



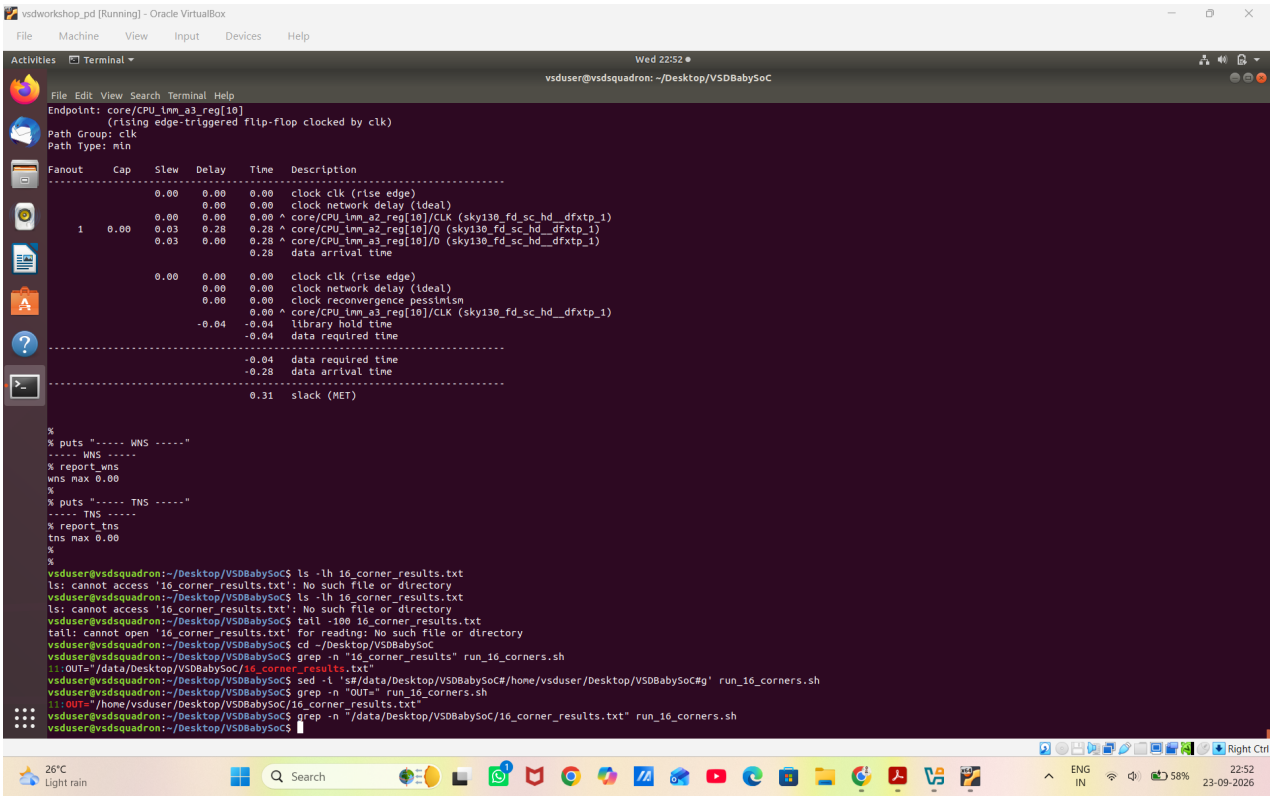
OpenSTA library/netlist linking stage showing the initial missing standard-cell definitions.

Evidence Screenshot 5



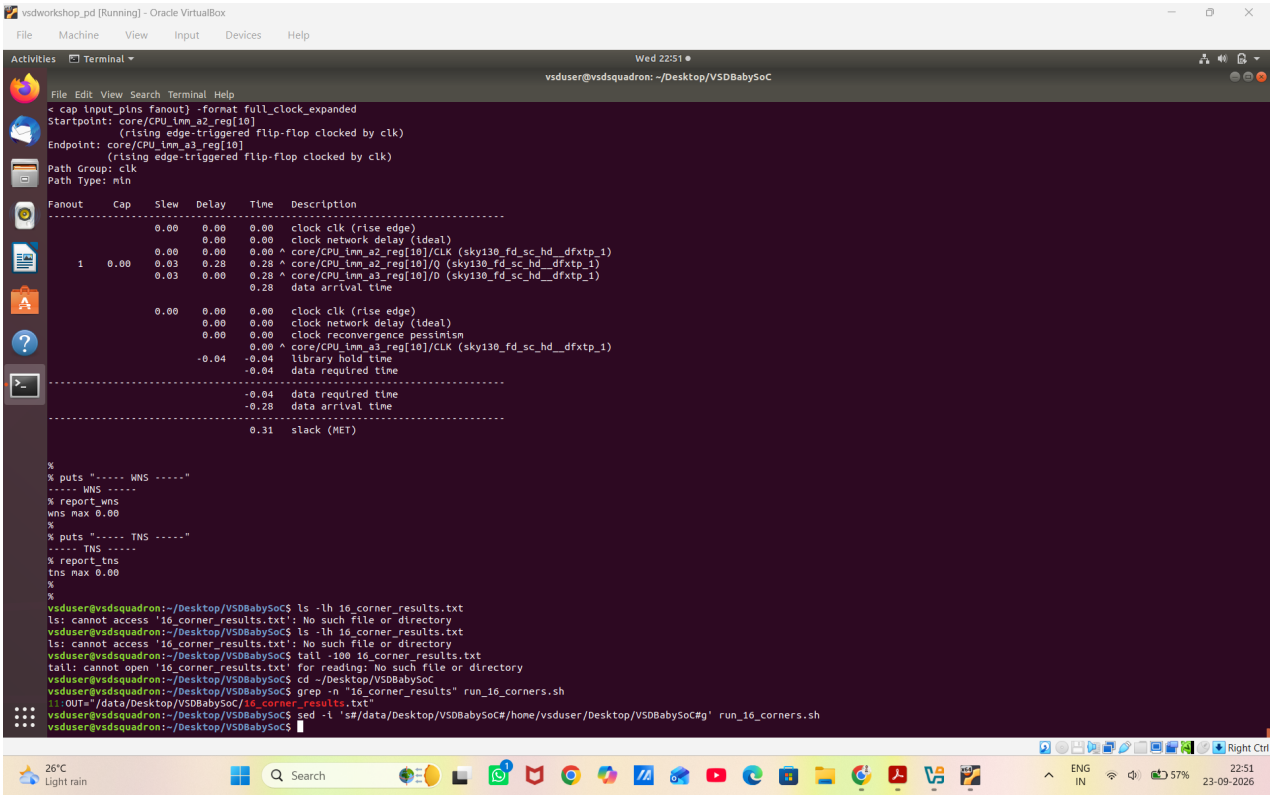
OpenSTA 3.1.0 startup and initial library/netlist loading evidence.

Evidence Screenshot 6



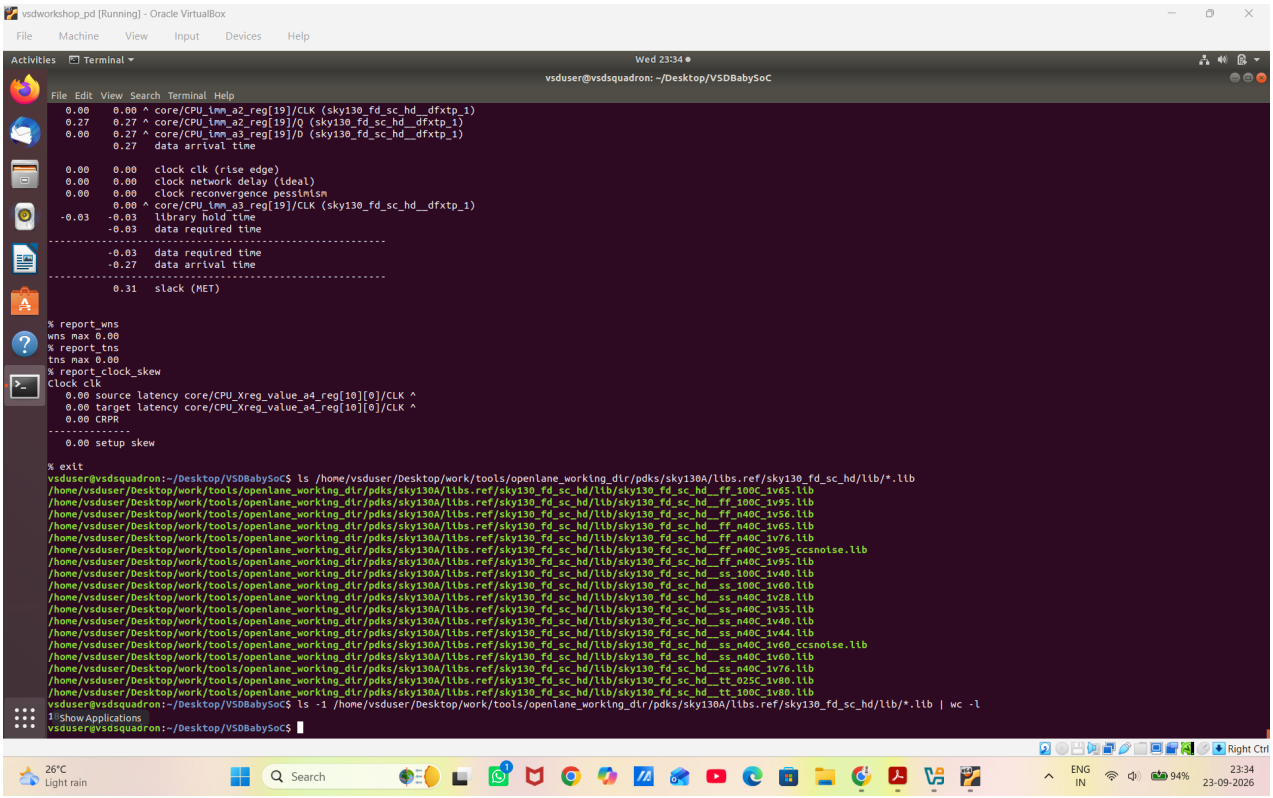
Timing-report attempt and subsequent terminal work during debugging.

Evidence Screenshot 7



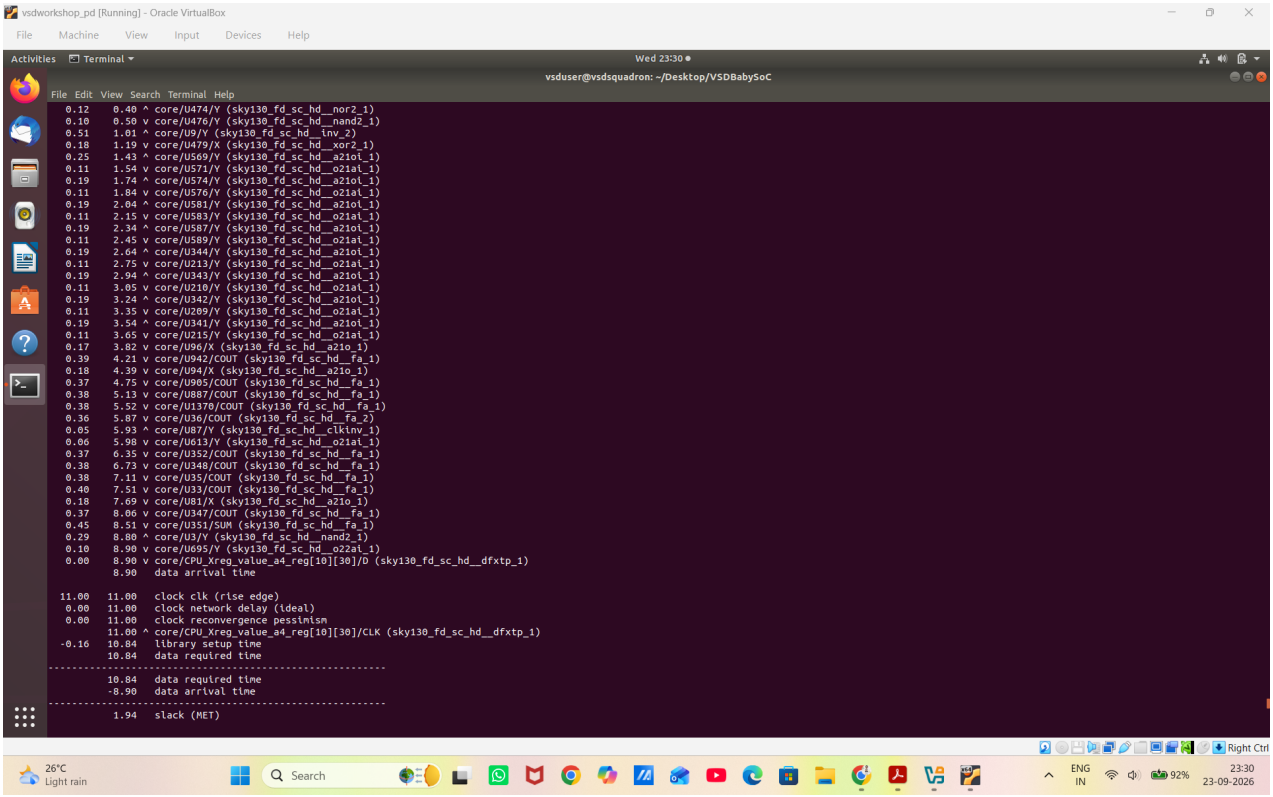
Additional terminal evidence from the timing-analysis/debugging sequence.

Evidence Screenshot 8



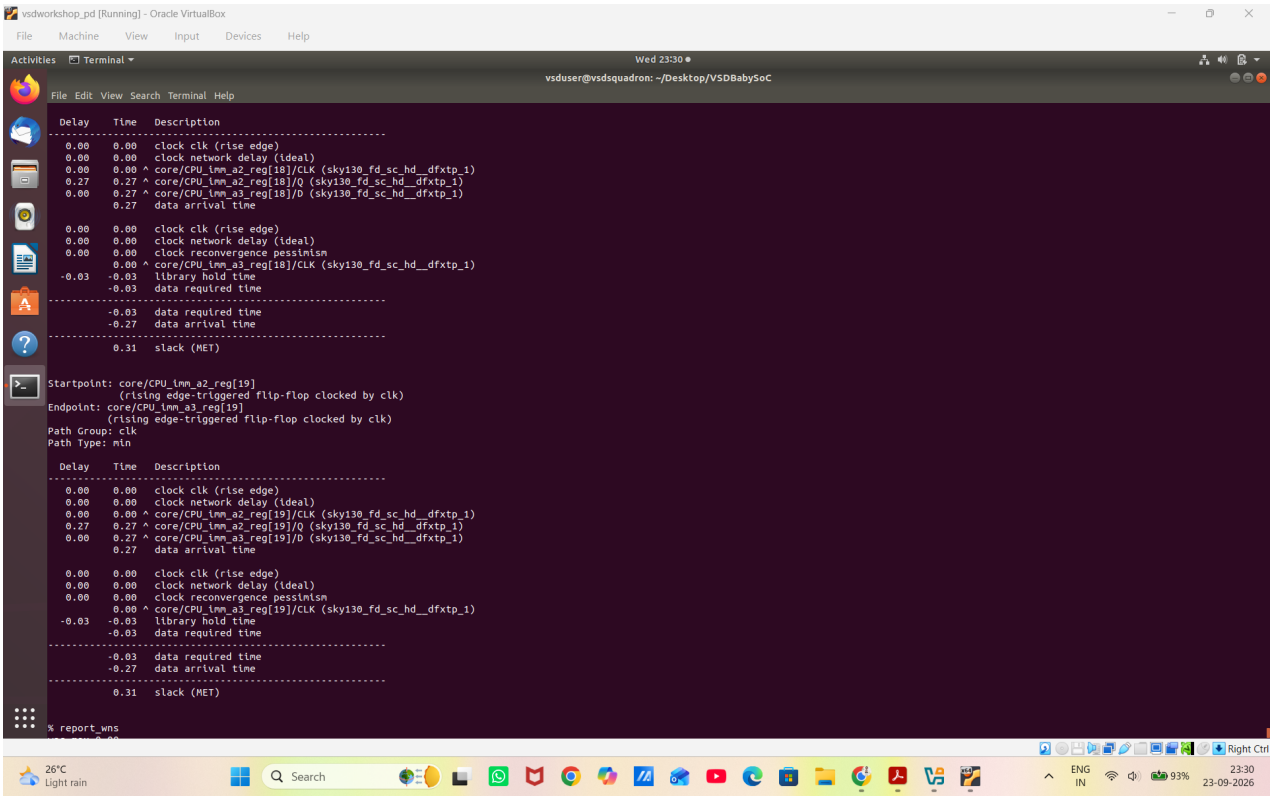
Listing of the available SKY130 HD corner Liberty files and count check.

Evidence Screenshot 9



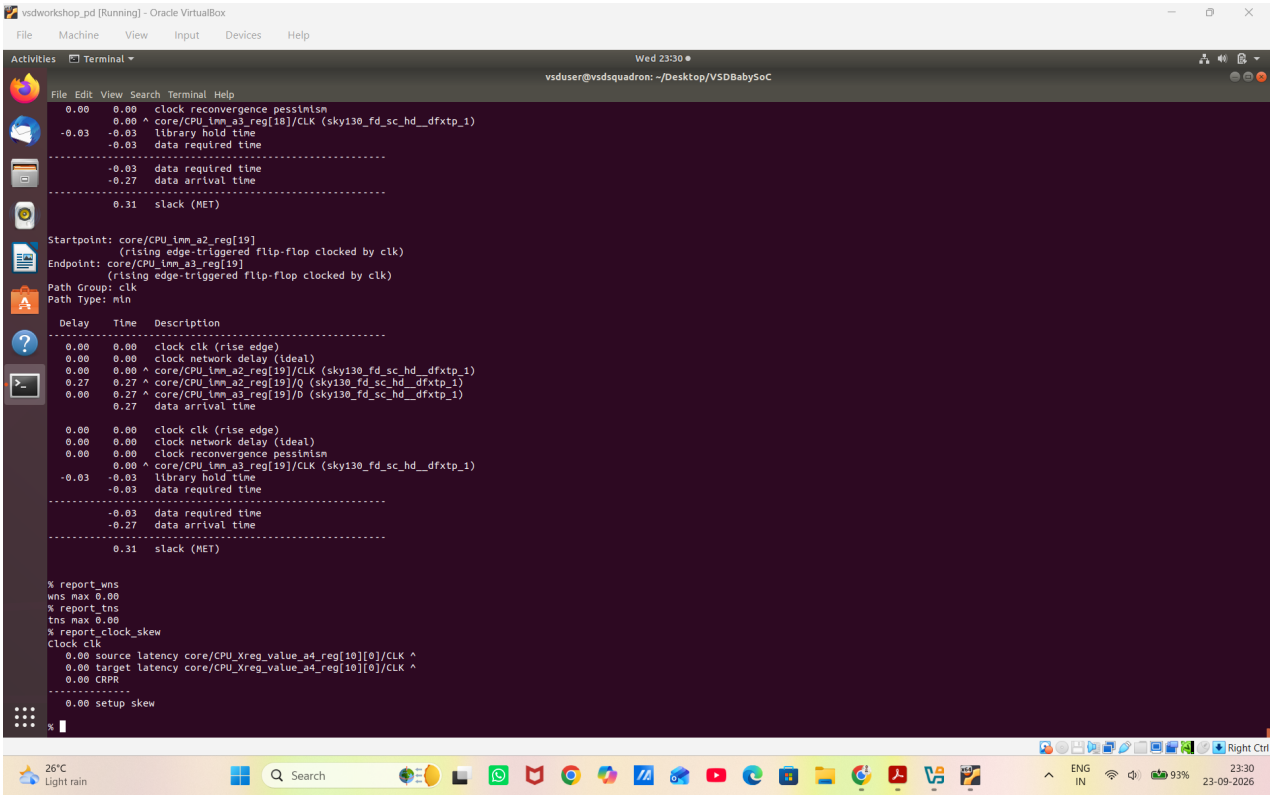
Representative setup path showing 8.90 ns arrival, 10.84 ns required time and +1.94 ns slack.

Evidence Screenshot 10



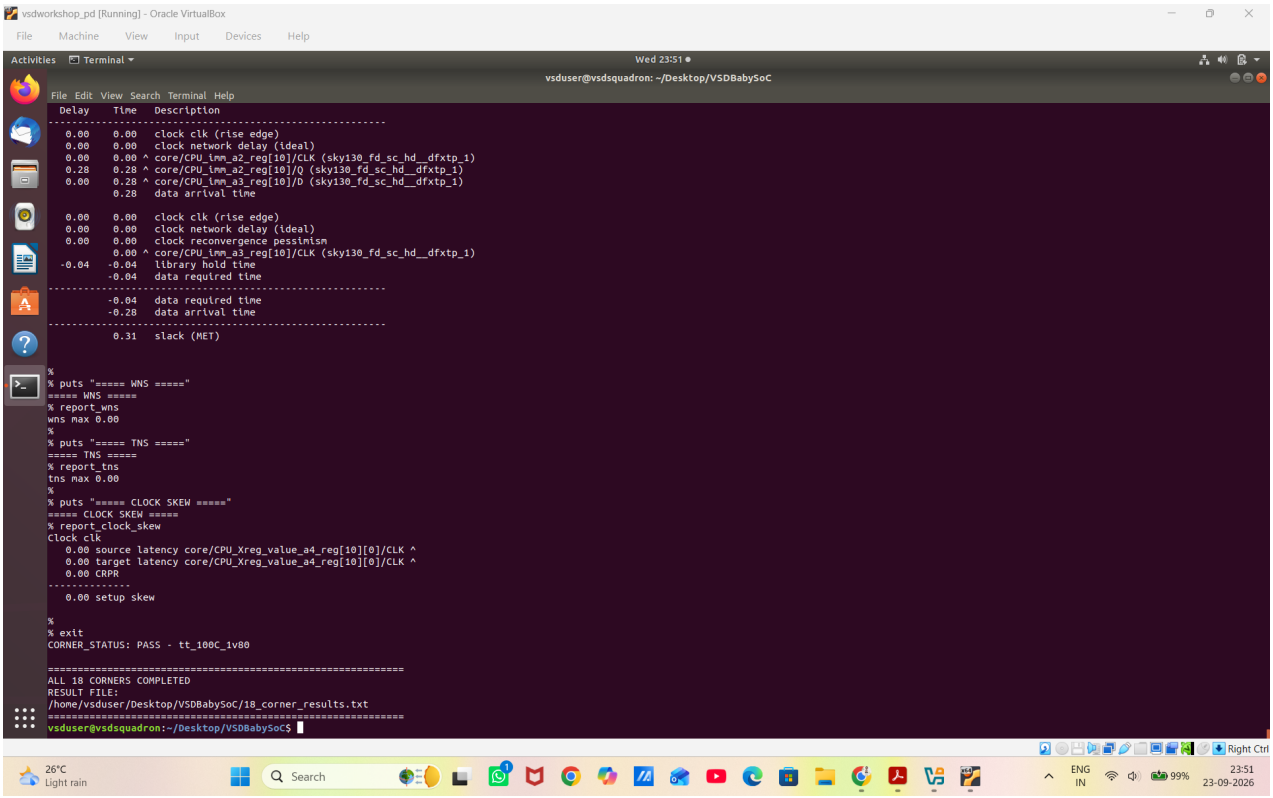
Representative hold/minimum-delay paths with positive +0.31 ns slack.

Evidence Screenshot 11



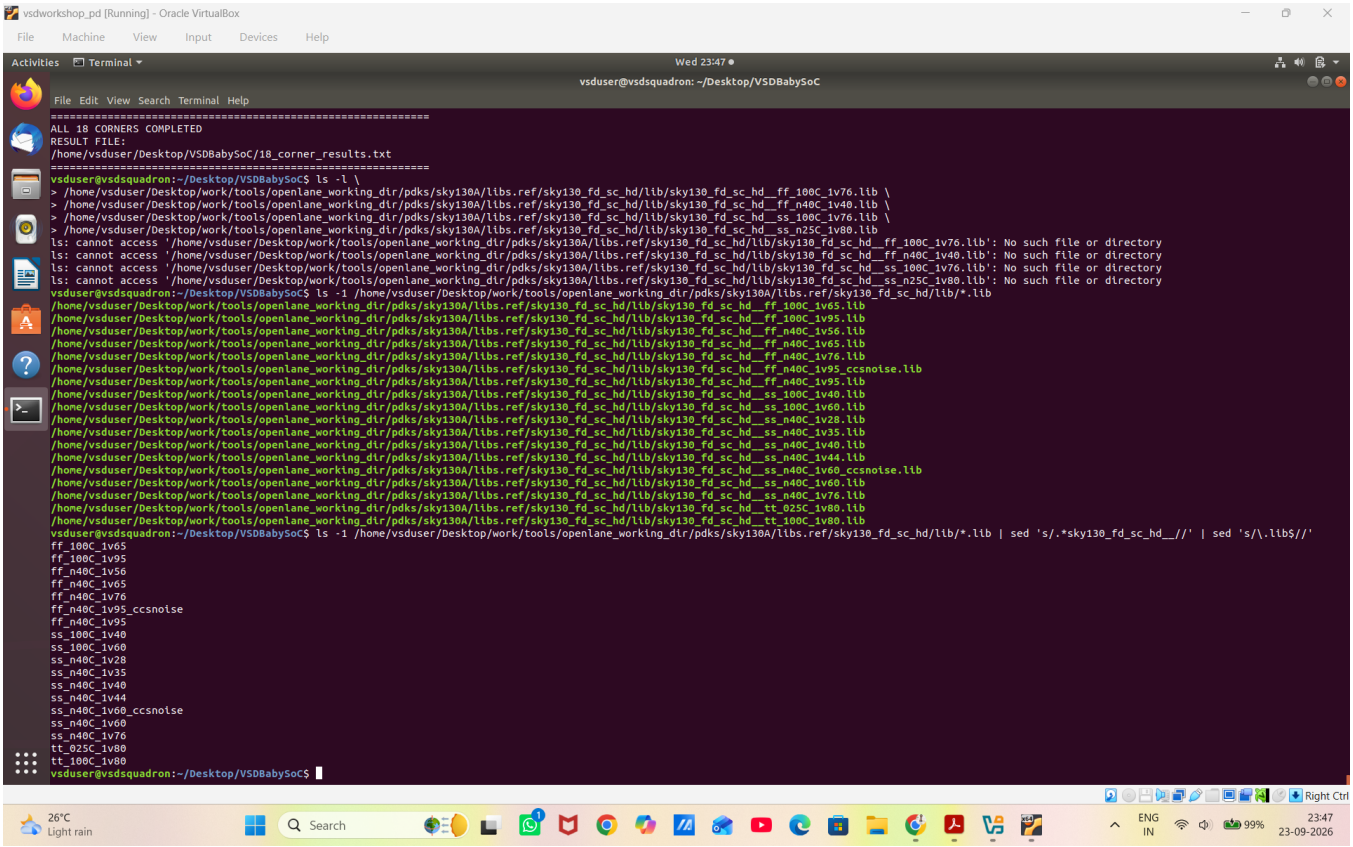
Hold result followed by WNS, TNS and clock-skew output.

Evidence Screenshot 12



Final terminal evidence: all 18 corners completed and the result file path was printed.

Evidence Screenshot 13



Final corner-library listing showing the 18 corner library names.

5. 18-Corner Results

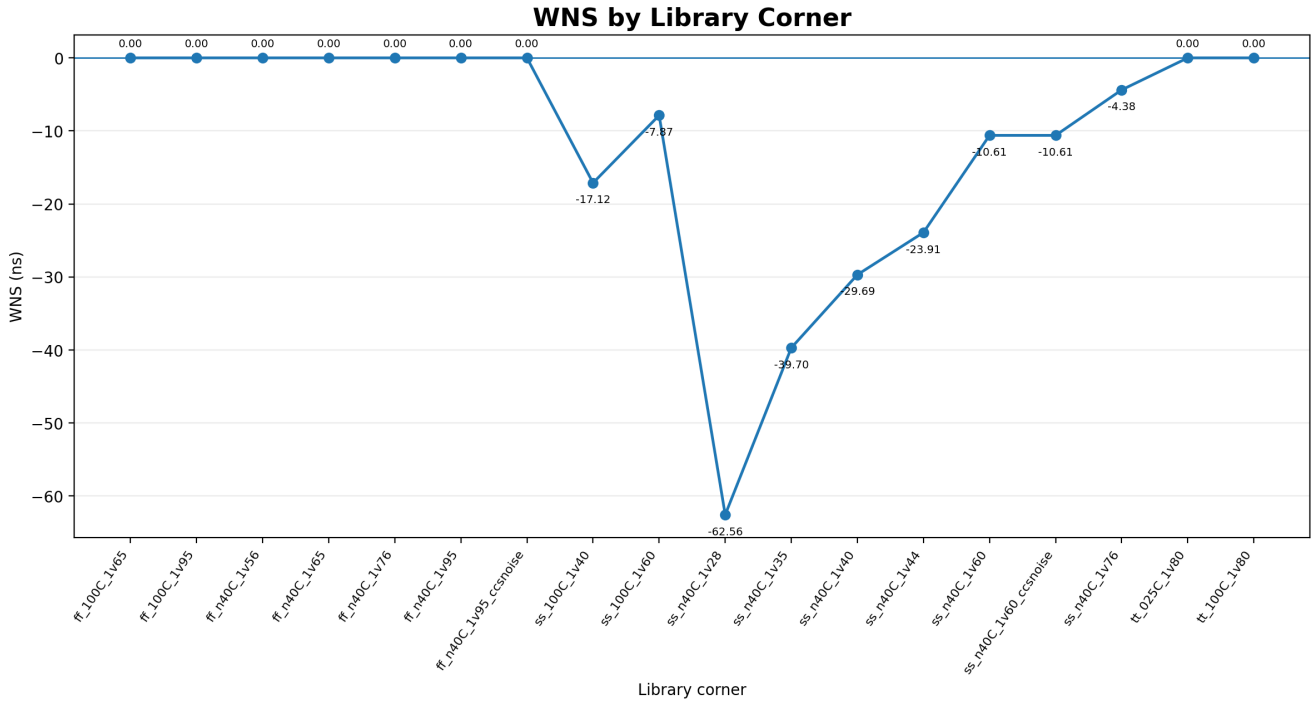
The final run covered the 18 cases represented by the SKY130 HD Liberty files visible in the terminal evidence. The two CCS-noise-labelled cases are retained as separate rows because they were part of the executed run.

Corner	P	Temp	V	Hold (ns)	WNS (ns)	TNS (ns)	Setup (ns)	Status
ff_100C_1v65	FF	100°C	1.65 V	0.25	0.00	0.00	4.05	MET
ff_100C_1v95	FF	100°C	1.95 V	0.20	0.00	0.00	5.57	MET
ff_n40C_1v56	FF	-40°C	1.56 V	0.29	0.00	0.00	2.28	MET
ff_n40C_1v65	FF	-40°C	1.65 V	0.26	0.00	0.00	3.41	MET
ff_n40C_1v76	FF	-40°C	1.76 V	0.22	0.00	0.00	4.43	MET
ff_n40C_1v95	FF	-40°C	1.95 V	0.19	0.00	0.00	5.59	MET
ff_n40C_1v95_ccsnoise	FF	-40°C	1.95 V	0.19	0.00	0.00	5.59	MET
ss_100C_1v40	**SS**	**100°C**	**1.40 V**	**0.91**	**-17.12**	**-2599.51**	**-17.12**	**VIOLATED**
ss_100C_1v60	**SS**	**100°C**	**1.60 V**	**0.64**	**-7.87**	**-892.70**	**-7.87**	**VIOLATED**
ss_n40C_1v28	**SS**	**-40°C**	**1.28 V**	**1.83**	**-62.56**	**-13957.59**	**-62.56**	**VIOLATED**
ss_n40C_1v35	**SS**	**-40°C**	**1.35 V**	**1.35**	**-39.70**	**-7993.79**	**-39.70**	**VIOLATED**
ss_n40C_1v40	**SS**	**-40°C**	**1.40 V**	**1.12**	**-29.69**	**-5466.48**	**-29.69**	**VIOLATED**
ss_n40C_1v44	**SS**	**-40°C**	**1.44 V**	**0.99**	**-23.91**	**-4107.19**	**-23.91**	**VIOLATED**
ss_n40C_1v60	**SS**	**-40°C**	**1.60 V**	**0.66**	**-10.61**	**-1355.80**	**-10.61**	**VIOLATED**
ss_n40C_1v60_ccsnoise	**SS**	**-40°C**	**1.60 V**	**0.66**	**-10.61**	**-1355.80**	**-10.61**	**VIOLATED**
ss_n40C_1v76	**SS**	**-40°C**	**1.76 V**	**0.50**	**-4.38**	**-367.93**	**-4.38**	**VIOLATED**
tt_025C_1v80	TT	25°C	1.80 V	0.31	0.00	0.00	1.94	MET
tt_100C_1v80	TT	100°C	1.80 V	0.31	0.00	0.00	2.09	MET

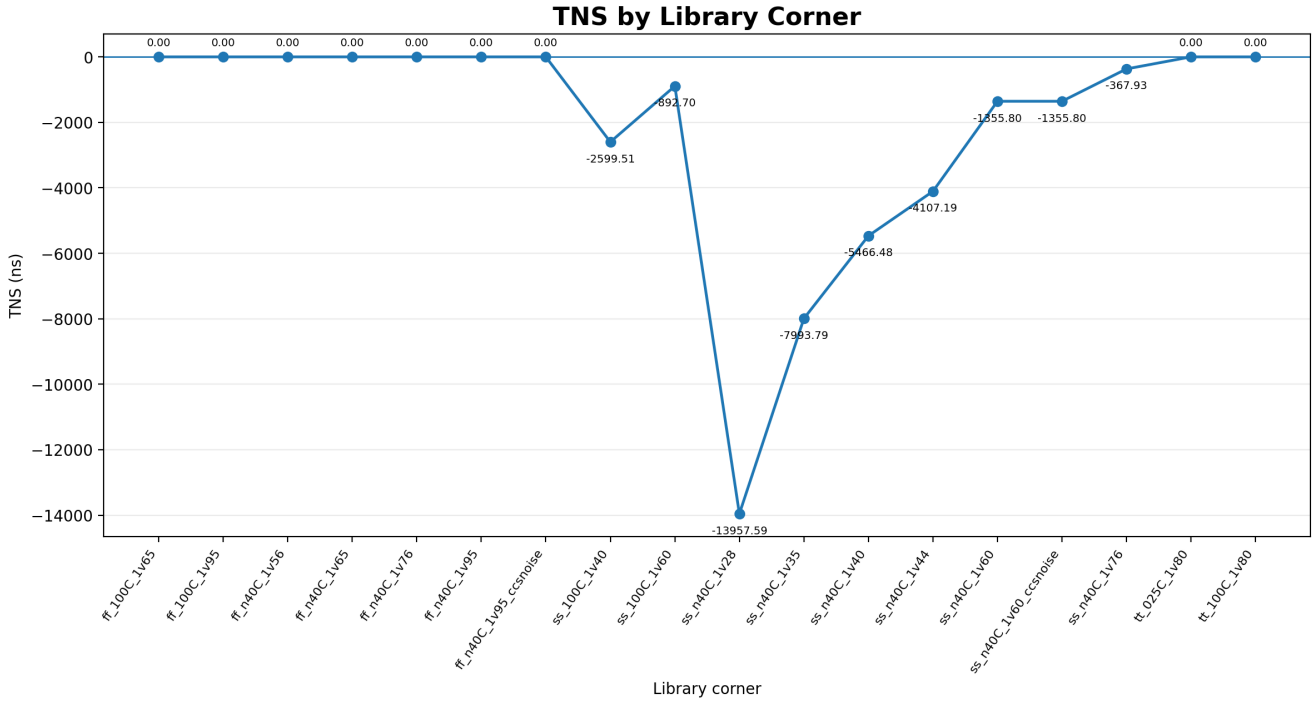
Note: for corners with no setup violation, OpenSTA's WNS/TNS summary is 0.00 ns, while the table retains the representative setup-path slack printed by the corner run. For violating SS corners, the reported setup-path slack matches the negative WNS shown in the run.

6. WNS and TNS Plots

WNS across the 18 recorded corners



TNS across the 18 recorded corners



7. Representative Timing Paths

Setup / maximum-delay path recorded in the working OpenSTA output:

```
Startpoint: core/CPU_is_addi_a3_reg
Endpoint: core/CPU_Xreg_value_a4_reg[10][30]
Path Group: clk
Path Type: max

... logic/cell propagation ...
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
           1.94   slack (MET)
```

Hold / minimum-delay path:

```
Startpoint: core/CPU_imm_a2_reg[10]
Endpoint: core/CPU_imm_a3_reg[10]
Path Group: clk
Path Type: min

   0.27   data arrival time
  -0.03   library hold time
          -0.03   data required time
-----------------------------------------------------------
           0.31   slack (MET)
```

Clock-skew output recorded for the representative run:

```
Clock clk
   0.00 source latency
   0.00 target latency
   0.00 CRPR
--------------
   0.00 setup skew
```

8. What the Results Show

The recorded FF and TT cases have non-negative setup-path slack in the final dataset. The SS cases show negative setup slack, with the largest recorded setup limitation at ss_n40C_1v28.

Item	Recorded result
Clock period	11.00 ns
Most negative WNS	-62.56 ns at ss_n40C_1v28
Most negative TNS	-13957.59 ns at ss_n40C_1v28
Hold slack at that corner	+1.83 ns
Representative normal setup path	+1.94 ns (MET)
Representative hold path	+0.31 ns (MET)
Final execution	ALL 18 CORNERS COMPLETED

PVT behavior in this run is limited to the synthesized design, 11 ns constraint and Liberty data used here. The SS cases in the lower-voltage conditions produced the negative setup results recorded in the table, while the FF cases retained positive setup-path slack.

9. Raw Terminal Output — Kept as Evidence

The following appendices reproduce the copied terminal-output files supplied from the working session. They are intentionally retained even where they contain warnings, intermediate attempts, or repeated OpenSTA startup text. This makes the report traceable to the actual work instead of only showing a polished final table.

Appendix A — Initial/representative OpenSTA timing output

Source file: Pasted text(5).txt

```
% get_cells -of_objects [get_pins core/CPU_Xreg_value_a4_reg[10][0]/CLK]
_d07d16a28c550000_p_Instance
% all_registers -clock [get_clocks clk]
_007215a28c550000_p_Instance _a08415a28c550000_p_Instance _80df15a28c550000_p_Instance _000d16a28c550000_p_Instance _001516a28c550000_p_Instance _
% report_checks -unconstrained -path_delay max -fields {slew cap input_pins fanout} -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.04    0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.00    0.28 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.11    0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                  0.11    0.00    0.40 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.08    0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                  0.08    0.00    0.50 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.15    0.67    0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                  0.67    0.00    1.01 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.17    0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                  0.17    0.00    1.19 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                  0.19    0.00    1.43 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.00    1.54 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                  0.19    0.00    1.74 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.00    1.84 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                  0.19    0.00    2.04 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.00    2.15 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
                  0.19    0.00    2.34 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.00    2.45 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
                  0.19    0.00    2.64 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.00    2.75 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
                  0.19    0.00    2.94 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.00    3.05 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
                  0.19    0.00    3.24 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.00    3.35 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
                  0.19    0.00    3.54 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
                  0.08    0.00    3.65 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.04    0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
                  0.04    0.00    3.82 v core/U942/CIN (sky130_fd_sc_hd__fa_1)
     2    0.01    0.09    0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
                  0.09    0.00    4.21 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.04    0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
                  0.04    0.00    4.39 v core/U905/CIN (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
                  0.08    0.00    4.75 v core/U887/CIN (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
                  0.08    0.00    5.13 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)
     1    0.01    0.08    0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
                  0.08    0.00    5.52 v core/U36/CIN (sky130_fd_sc_hd__fa_2)
     2    0.01    0.07    0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
                  0.07    0.00    5.87 v core/U87/A (sky130_fd_sc_hd__clkinv_1)
     1    0.00    0.03    0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
                  0.03    0.00    5.93 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)
     1    0.00    0.06    0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
                  0.06    0.00    5.98 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
                  0.08    0.00    6.35 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
                  0.08    0.00    6.73 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
                  0.08    0.00    7.11 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
     2    0.01    0.09    0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
                  0.09    0.00    7.51 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
     1    0.00    0.04    0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
                  0.04    0.00    7.69 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
     1    0.00    0.08    0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
                  0.08    0.00    8.06 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
     2    0.01    0.09    0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
                  0.09    0.00    8.51 v core/U3/A (sky130_fd_sc_hd__nand2_1)
    15    0.04    0.35    0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
```

```
                0.35    0.00    8.80 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
     1    0.00  0.14    0.10    8.90 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
                0.14    0.00    8.90 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
                                8.90   data arrival time

                0.00   11.00   11.00   clock clk (rise edge)
                        0.00   11.00   clock network delay (ideal)
                        0.00   11.00   clock reconvergence pessimism
                               11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
                       -0.16   10.84   library setup time
                               10.84   data required time
-----------------------------------------------------------------------------
                               10.84   data required time
```

Appendix A — Initial/representative OpenSTA timing output — continued (lines 106 onward)

```
                                -8.90   data arrival time
---------------------------------------------------------------------------
                                 1.94   slack (MET)


%
```

Appendix B — Detailed setup and hold report output

Source file: Pasted text(6).txt

```
% report_checks -path_delay max -group_count 10 -format full_clock_expanded
Warning 503: vsdbabysoc_synthesis.sdc line 266, report_checks -group_count is deprecated. Use -group_path_count instead.
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
---------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
---------------------------------------------------------
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[19][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

```
0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
```

Appendix B — Detailed setup and hold report output — continued (lines 106 onward)

```
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U692/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[19][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[19][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[26][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U691/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[26][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[26][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
```

```
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[11][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

Appendix B — Detailed setup and hold report output — continued (lines 211 onward)

```
  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U687/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[11][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[11][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
---------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
---------------------------------------------------------
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[25][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
```

```
0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
```

Appendix B — Detailed setup and hold report output — continued (lines 316 onward)

```
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U689/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[25][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[25][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[9][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U686/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[9][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[9][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[18][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

```
   Delay    Time   Description
---------------------------------------------------------
    0.00    0.00   clock clk (rise edge)
    0.00    0.00   clock network delay (ideal)
    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
    0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
    0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
    0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
    0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
```

Appendix B — Detailed setup and hold report output — continued (lines 421 onward)

```
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U694/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[18][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[18][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[17][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
```

```
  0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
  0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
  0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
  0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
  0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
  0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
  0.10    8.90 v core/U698/Y (sky130_fd_sc_hd__o22ai_1)
  0.00    8.90 v core/CPU_Xreg_value_a4_reg[17][30]/D (sky130_fd_sc_hd__dfxtp_1)
          8.90   data arrival time
```

Appendix B — Detailed setup and hold report output — continued (lines 526 onward)

```
  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[17][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[3][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U688/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[3][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[3][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
           1.94   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[10][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
```

```
0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
```

Appendix B — Detailed setup and hold report output — continued (lines 631 onward)

```
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.10    8.90 v core/U695/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.90 v core/CPU_Xreg_value_a4_reg[10][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.90   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[10][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
           1.94   slack (MET)


% report_checks -path_delay min -group_count 10 -format full_clock_expanded
Warning 503: vsdbabysoc_synthesis.sdc line 266, report_checks -group_count is deprecated. Use -group_path_count instead.
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
---------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
---------------------------------------------------------
           0.31   slack (MET)


Startpoint: core/CPU_imm_a2_reg[11]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[11]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[11]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[11]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[11]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[11]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
```

```
-----------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
-----------------------------------------------------------
           0.31   slack (MET)


Startpoint: core/CPU_imm_a2_reg[12]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[12]
```

Appendix B — Detailed setup and hold report output — continued (lines 736 onward)

```
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[12]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[12]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[12]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[12]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
---------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
---------------------------------------------------------
           0.31   slack (MET)


Startpoint: core/CPU_imm_a2_reg[13]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[13]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[13]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[13]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[13]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[13]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
---------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
---------------------------------------------------------
           0.31   slack (MET)


Startpoint: core/CPU_imm_a2_reg[14]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[14]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[14]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[14]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[14]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[14]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
---------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
---------------------------------------------------------
           0.31   slack (MET)


Startpoint: core/CPU_imm_a2_reg[15]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[15]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
```

```
   0.00    0.00 ^ core/CPU_imm_a2_reg[15]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[15]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[15]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[15]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
```

Appendix B — Detailed setup and hold report output — continued (lines 841 onward)

```
          -0.03   data required time
-----------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
-----------------------------------------------------------
           0.31   slack (MET)


Startpoint: core/CPU_imm_a2_reg[16]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[16]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[16]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[16]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[16]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[16]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
-----------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
-----------------------------------------------------------
           0.31   slack (MET)


Startpoint: core/CPU_imm_a2_reg[17]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[17]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[17]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[17]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[17]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[17]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
-----------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
-----------------------------------------------------------
           0.31   slack (MET)


Startpoint: core/CPU_imm_a2_reg[18]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[18]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[18]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[18]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[18]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[18]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
-----------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
-----------------------------------------------------------
           0.31   slack (MET)

Startpoint: core/CPU_imm_a2_reg[19]
```

```
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[19]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
```

Appendix B — Detailed setup and hold report output — continued (lines 946 onward)

```
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[19]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[19]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[19]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[19]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
-----------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
-----------------------------------------------------------
           0.31   slack (MET)


% report_wns
wns max 0.00
% report_tns
tns max 0.00
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
```

Appendix D — Full 18-corner terminal run output

Source file: Pasted text(8).txt

```
vsduser@vsdsquadron:~/Desktop/VSDBabySoC$ ./run_all_18_corners.sh

============================================================
RUNNING CORNER: ff_100C_1v65
============================================================

############################################################
CORNER: ff_100C_1v65
LIBRARY: sky130_fd_sc_hd__ff_100C_1v65.lib
############################################################

OpenSTA 3.1.0 2996e37a3a Copyright (c) 2026, Parallax Software, Inc.
License GPLv3: GNU GPL version 3 <http://gnu.org/licenses/gpl.html>

This is free software, and you are free to change and redistribute it
under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_100C_1v65.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_100C_1v65.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ff_100C_1v65 ====="
===== CORNER: ff_100C_1v65 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.24    0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.09    0.34 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.08    0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.42    0.83 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.13    0.96 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.19    1.16 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    1.25 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.15    1.40 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    1.49 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.15    1.64 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    1.73 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.15    1.89 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    1.98 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.15    2.13 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    2.22 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.15    2.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    2.46 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.15    2.61 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    2.71 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.15    2.86 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    2.95 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.14    3.09 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.28    3.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.14    3.50 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.26    3.76 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.27    4.03 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.27    4.31 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.28    4.59 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.04    4.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.05    4.68 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.26    4.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.27    5.21 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.27    5.48 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.29    5.77 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.14    5.91 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.26    6.17 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.55 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.20    6.75 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.13    6.88 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
```

```
   0.00    6.88 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
           6.88   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[16][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.07   10.93   library setup time
          10.93   data required time
-----------------------------------------------------------
          10.93   data required time
          -6.88   data arrival time
-----------------------------------------------------------
```

Appendix D — Full 18-corner terminal run output — continued (lines 106 onward)

```
           4.05   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.22    0.22 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.22 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.22   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
---------------------------------------------------------
          -0.03   data required time
          -0.22   data arrival time
---------------------------------------------------------
           0.25   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ff_100C_1v65

============================================================
RUNNING CORNER: ff_100C_1v95
============================================================

############################################################
CORNER: ff_100C_1v95
LIBRARY: sky130_fd_sc_hd__ff_100C_1v95.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_100C_1v95.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_100C_1v95.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ff_100C_1v95 ====="
```

```
===== CORNER: ff_100C_1v95 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
```

Appendix D — Full 18-corner terminal run output — continued (lines 211 onward)

```
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.20    0.20 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.07    0.27 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.07    0.33 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.35    0.68 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.09    0.77 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.15    0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    0.99 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.12    1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.18 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.12    1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.12    1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.12    1.67 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.12    1.86 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.12    2.05 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.12    2.24 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    2.31 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.11    2.42 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.22    2.63 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.11    2.74 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.20    2.95 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.21    3.16 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.21    3.37 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.22    3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.03    3.61 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.04    3.66 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.21    4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.21    4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.22    4.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.11    4.62 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.20    4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.30    5.12 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.17    5.28 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.11    5.39 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    5.39 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
           5.39   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[16][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.04   10.96   library setup time
          10.96   data required time
---------------------------------------------------------
          10.96   data required time
          -5.39   data arrival time
---------------------------------------------------------
           5.57   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.18    0.18 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.18 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.18   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.02   -0.02   library hold time
          -0.02   data required time
---------------------------------------------------------
          -0.02   data required time
          -0.18   data arrival time
---------------------------------------------------------
           0.20   slack (MET)

%
```

```
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
```

Appendix D — Full 18-corner terminal run output — continued (lines 316 onward)

```
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ff_100C_1v95

============================================================
RUNNING CORNER: ff_n40C_1v56
============================================================

############################################################
CORNER: ff_n40C_1v56
LIBRARY: sky130_fd_sc_hd__ff_n40C_1v56.lib
############################################################

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This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v56.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v56.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ff_n40C_1v56 ====="
===== CORNER: ff_n40C_1v56 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.39 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.08    0.48 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.54    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.15    1.16 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.27    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.10    1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.10    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.10    2.14 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    2.35 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.10    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    2.65 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.10    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    2.96 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.10    3.06 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    3.26 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.10    3.36 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    3.57 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.10    3.67 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.84 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.35    4.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.38 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.33    4.71 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.35    5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.35    5.41 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.34    5.74 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.06    5.80 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
```

```
0.05    5.85 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.34    6.18 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.35    6.53 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.35    6.88 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.36    7.24 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.18    7.42 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.33    7.76 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.34    8.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.22    8.32 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.16    8.48 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
```

Appendix D — Full 18-corner terminal run output — continued (lines 421 onward)

```
   0.16    8.64 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.64 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
           8.64   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[16][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.08   10.92   library setup time
          10.92   data required time
-----------------------------------------------------------
          10.92   data required time
          -8.64   data arrival time
-----------------------------------------------------------
           2.28   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.26    0.26 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.26 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.26   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
---------------------------------------------------------
          -0.03   data required time
          -0.26   data arrival time
---------------------------------------------------------
           0.29   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ff_n40C_1v56

============================================================
RUNNING CORNER: ff_n40C_1v65
============================================================

############################################################
CORNER: ff_n40C_1v65
LIBRARY: sky130_fd_sc_hd__ff_n40C_1v65.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
```

```
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v65.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v65.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
```

Appendix D — Full 18-corner terminal run output — continued (lines 526 onward)

```
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ff_n40C_1v65 ====="
===== CORNER: ff_n40C_1v65 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.24    0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.11    0.35 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.07    0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.48    0.90 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.12    1.02 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.23    1.26 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    1.34 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    1.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    1.61 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    1.79 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    1.88 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    2.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    2.15 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    2.33 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    2.41 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    2.59 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    2.68 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    2.86 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    2.95 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    3.13 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.09    3.22 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.15    3.37 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.30    3.67 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.16    3.83 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.29    4.12 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.30    4.42 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.30    4.72 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.29    5.01 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.06 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.04    5.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.29    5.39 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.30    5.69 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.30    5.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.31    6.30 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.16    6.46 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.29    6.75 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.30    7.05 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
   0.19    7.24 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
   0.14    7.38 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
   0.14    7.52 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    7.52 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
           7.52   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[16][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.07   10.93   library setup time
          10.93   data required time
---------------------------------------------------------
          10.93   data required time
          -7.52   data arrival time
---------------------------------------------------------
           3.41   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.23    0.23 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.23 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
```

```
           0.23   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.02   -0.02   library hold time
          -0.02   data required time
-----------------------------------------------------------
          -0.02   data required time
```

Appendix D — Full 18-corner terminal run output — continued (lines 631 onward)

```
          -0.23   data arrival time
-----------------------------------------------------------
           0.26   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ff_n40C_1v65

============================================================
RUNNING CORNER: ff_n40C_1v76
============================================================

############################################################
CORNER: ff_n40C_1v76
LIBRARY: sky130_fd_sc_hd__ff_n40C_1v76.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v76.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v76.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ff_n40C_1v76 ====="
===== CORNER: ff_n40C_1v76 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[1][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.21    0.21 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.10    0.31 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.06    0.37 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.43    0.81 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.09    0.90 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.20    1.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.17 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.16    1.33 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.41 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.16    1.56 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.64 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.16    1.80 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.07    1.87 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.16    2.03 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
```

```
0.07    2.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.16    2.26 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.07    2.33 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.16    2.49 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.07    2.56 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.16    2.72 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.07    2.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.13    2.93 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.26    3.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.14    3.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)
```

Appendix D — Full 18-corner terminal run output — continued (lines 736 onward)

```
   0.25    3.58 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.26    3.83 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.26    4.09 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.25    4.34 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.04    4.38 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.04    4.42 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.25    4.68 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.26    4.93 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.26    5.19 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.27    5.46 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.14    5.60 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.25    5.85 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.25    6.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
   0.17    6.27 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
   0.12    6.39 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
   0.12    6.52 ^ core/U643/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    6.52 ^ core/CPU_Xreg_value_a4_reg[1][31]/D (sky130_fd_sc_hd__dfxtp_1)
           6.52   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[1][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.05   10.95   library setup time
          10.95   data required time
-----------------------------------------------------------
          10.95   data required time
          -6.52   data arrival time
-----------------------------------------------------------
           4.43   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.20    0.20 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.20 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.20   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.02   -0.02   library hold time
          -0.02   data required time
---------------------------------------------------------
          -0.02   data required time
          -0.20   data arrival time
---------------------------------------------------------
           0.22   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ff_n40C_1v76

============================================================
RUNNING CORNER: ff_n40C_1v95
============================================================

############################################################
```

```
CORNER: ff_n40C_1v95
LIBRARY: sky130_fd_sc_hd__ff_n40C_1v95.lib
############################################################

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```

Appendix D — Full 18-corner terminal run output — continued (lines 841 onward)

```
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v95.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v95.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ff_n40C_1v95 ====="
===== CORNER: ff_n40C_1v95 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.18    0.18 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.08    0.26 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.05    0.32 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.37    0.69 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.06    0.75 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.17    0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    0.98 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.17 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    1.68 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    1.87 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    2.06 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    2.26 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    2.32 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.11    2.43 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.21    2.64 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.11    2.75 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.20    2.96 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.21    3.17 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.21    3.38 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.20    3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.03    3.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.04    3.65 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.21    4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.21    4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.22    4.50 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.11    4.61 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.20    4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.21    5.02 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
   0.14    5.17 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
   0.10    5.27 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
   0.10    5.37 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    5.37 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
           5.37   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[16][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.04   10.96   library setup time
          10.96   data required time
---------------------------------------------------------
          10.96   data required time
          -5.37   data arrival time
---------------------------------------------------------
           5.59   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
```

```
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
```

Appendix D — Full 18-corner terminal run output — continued (lines 946 onward)

```
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.17    0.17 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.17 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.17   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.02   -0.02   library hold time
          -0.02   data required time
-----------------------------------------------------------
          -0.02   data required time
          -0.17   data arrival time
-----------------------------------------------------------
           0.19   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ff_n40C_1v95

============================================================
RUNNING CORNER: ff_n40C_1v95_ccsnoise
============================================================

############################################################
CORNER: ff_n40C_1v95_ccsnoise
LIBRARY: sky130_fd_sc_hd__ff_n40C_1v95_ccsnoise.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<c_hd/lib/sky130_fd_sc_hd__ff_n40C_1v95_ccsnoise.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ff_n40C_1v95_ccsnoise.lib l
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ff_n40C_1v95_ccsnoise ====="
===== CORNER: ff_n40C_1v95_ccsnoise =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
```

```
0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.18    0.18 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.08    0.26 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.05    0.32 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.37    0.69 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.06    0.75 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.17    0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.06    0.98 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.13    1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.06    1.17 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
```

Appendix D — Full 18-corner terminal run output — continued (lines 1051 onward)

```
   0.13    1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    1.68 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    1.87 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    2.06 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.13    2.26 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.06    2.32 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.11    2.43 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.21    2.64 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.11    2.75 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.20    2.96 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.21    3.17 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.21    3.38 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.20    3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.03    3.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.04    3.65 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.21    3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.21    4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.21    4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.22    4.50 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.11    4.61 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.20    4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.21    5.02 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
   0.14    5.17 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
   0.10    5.27 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
   0.10    5.37 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    5.37 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
           5.37   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[16][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.04   10.96   library setup time
          10.96   data required time
-----------------------------------------------------------
          10.96   data required time
          -5.37   data arrival time
-----------------------------------------------------------
           5.59   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.17    0.17 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.17 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.17   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.02   -0.02   library hold time
          -0.02   data required time
-----------------------------------------------------------
          -0.02   data required time
          -0.17   data arrival time
-----------------------------------------------------------
           0.19   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
```

```
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ff_n40C_1v95_ccsnoise
```

Appendix D — Full 18-corner terminal run output — continued (lines 1156 onward)

```
============================================================
RUNNING CORNER: ss_100C_1v40
============================================================

############################################################
CORNER: ss_100C_1v40
LIBRARY: sky130_fd_sc_hd__ss_100C_1v40.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_100C_1v40.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_100C_1v40.lib line 24, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_100C_1v40 ====="
===== CORNER: ss_100C_1v40 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.21    1.13 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
   0.18    1.30 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
   0.71    2.01 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
   0.80    2.81 v core/U478/X (sky130_fd_sc_hd__a22o_1)
   0.47    3.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.58    3.86 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.36    4.22 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.48    4.70 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.36    5.05 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.48    5.53 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.36    5.89 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.48    6.37 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.36    6.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.48    7.20 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.36    7.56 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.48    8.04 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.36    8.39 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.48    8.87 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.36    9.23 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.48    9.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.36   10.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.49   10.55 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   1.32   11.87 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.51   12.38 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   1.27   13.65 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   1.33   14.98 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   1.34   16.32 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   1.13   17.45 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.13   17.58 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.17   17.75 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   1.30   19.05 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   1.33   20.38 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   1.33   21.70 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   1.37   23.08 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.51   23.59 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   1.27   24.86 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   1.63   26.49 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.66   27.15 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.39   27.54 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00   27.54 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          27.54   data arrival time

  11.00   11.00   clock clk (rise edge)
```

```
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.58   10.42   library setup time
          10.42   data required time
-----------------------------------------------------------
          10.42   data required time
         -27.54   data arrival time
-----------------------------------------------------------
         -17.12   slack (VIOLATED)
```

Appendix D — Full 18-corner terminal run output — continued (lines 1261 onward)

```
%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.75    0.75 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.75 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.75   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.15   -0.15   library hold time
          -0.15   data required time
---------------------------------------------------------
          -0.15   data required time
          -0.75   data arrival time
---------------------------------------------------------
           0.91   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -17.12
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max -2599.51
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ss_100C_1v40

============================================================
RUNNING CORNER: ss_100C_1v60
============================================================

############################################################
CORNER: ss_100C_1v60
LIBRARY: sky130_fd_sc_hd__ss_100C_1v60.lib
############################################################

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This is free software, and you are free to change and redistribute it
under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_100C_1v60.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_100C_1v60.lib line 24, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_100C_1v60 ====="
===== CORNER: ss_100C_1v60 =====
%
% puts "===== SETUP (MAX) ====="
```

```
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

Appendix D — Full 18-corner terminal run output — continued (lines 1366 onward)

```
  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.62    0.62 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.15    0.78 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
   0.13    0.91 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.42 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
   0.56    1.98 v core/U478/X (sky130_fd_sc_hd__a22o_1)
   0.33    2.31 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.42    2.72 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    2.97 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.34    3.31 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    3.55 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.34    3.89 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    4.13 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.34    4.47 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    4.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.34    5.06 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    5.30 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.34    5.64 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    5.88 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.34    6.22 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    6.46 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.34    6.81 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    7.05 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.34    7.38 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.86    8.24 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.35    8.59 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.83    9.42 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.86   10.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.87   11.14 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.74   11.89 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.09   11.98 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.12   12.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.84   12.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.86   13.80 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.86   14.66 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.89   15.55 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.35   15.90 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.83   16.73 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   1.04   17.77 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.48   18.25 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.26   18.51 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00   18.51 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          18.51   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.36   10.64   library setup time
          10.64   data required time
---------------------------------------------------------
          10.64   data required time
         -18.51   data arrival time
---------------------------------------------------------
          -7.87   slack (VIOLATED)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
---------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
---------------------------------------------------------
           0.64   slack (MET)


%
```

```
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -7.87
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max -892.70
```

Appendix D — Full 18-corner terminal run output — continued (lines 1471 onward)

```
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ss_100C_1v60

============================================================
RUNNING CORNER: ss_n40C_1v28
============================================================

############################################################
CORNER: ss_n40C_1v28
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v28.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v28.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v28.lib line 24, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_n40C_1v28 ====="
===== CORNER: ss_n40C_1v28 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.50    1.50 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.71    2.20 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.56    2.76 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   2.67    5.43 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   1.71    7.14 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   1.42    8.57 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.76    9.33 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   1.20   10.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.76   11.29 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   1.20   12.49 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.76   13.25 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   1.20   14.45 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.76   15.21 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   1.20   16.41 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.76   17.17 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   1.20   18.38 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.76   19.14 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   1.20   20.34 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.76   21.10 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   1.20   22.30 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.76   23.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   1.13   24.19 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   3.70   27.89 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   1.32   29.21 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   3.62   32.83 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   3.78   36.61 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   3.80   40.42 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   3.39   43.81 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.40   44.21 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
```

```
0.21   44.43 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
3.63   48.06 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
3.78   51.84 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
3.78   55.62 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
3.86   59.48 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
1.32   60.80 v core/U81/X (sky130_fd_sc_hd__a21o_1)
3.62   64.42 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
4.76   69.19 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
1.61   70.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.91   71.70 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
```

Appendix D — Full 18-corner terminal run output — continued (lines 1576 onward)

```
   0.00   71.70 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          71.70   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.86    9.14   library setup time
           9.14   data required time
-----------------------------------------------------------
           9.14   data required time
         -71.70   data arrival time
-----------------------------------------------------------
         -62.56   slack (VIOLATED)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.45    1.45 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    1.45 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           1.45   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.38   -0.38   library hold time
          -0.38   data required time
---------------------------------------------------------
          -0.38   data required time
          -1.45   data arrival time
---------------------------------------------------------
           1.83   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -62.56
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max -13957.59
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ss_n40C_1v28

============================================================
RUNNING CORNER: ss_n40C_1v35
============================================================

############################################################
CORNER: ss_n40C_1v35
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v35.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
```

```
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v35.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v35.lib line 24, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
```

Appendix D — Full 18-corner terminal run output — continued (lines 1681 onward)

```
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_n40C_1v35 ====="
===== CORNER: ss_n40C_1v35 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.12    1.12 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.52    1.64 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.41    2.05 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   2.01    4.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   1.22    5.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   1.04    6.33 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.54    6.87 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.87    7.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.54    8.28 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.87    9.15 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.54    9.69 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.87   10.56 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.54   11.10 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.87   11.96 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.54   12.51 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.87   13.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.54   13.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.87   14.78 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.54   15.33 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.87   16.19 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.54   16.74 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.81   17.54 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   2.47   20.01 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.91   20.92 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   2.42   23.34 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   2.52   25.87 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   2.54   28.40 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   2.27   30.67 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.27   30.94 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.17   31.11 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   2.43   33.54 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   2.52   36.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   2.52   38.59 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   2.57   41.16 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.91   42.07 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   2.42   44.49 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   3.14   47.63 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   1.20   48.83 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.64   49.47 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00   49.47 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          49.47   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.23    9.77   library setup time
           9.77   data required time
---------------------------------------------------------
           9.77   data required time
         -49.47   data arrival time
---------------------------------------------------------
         -39.70   slack (VIOLATED)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.08    1.08 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    1.08 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           1.08   data arrival time
```

```
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.27   -0.27   library hold time
          -0.27   data required time
-----------------------------------------------------------
          -0.27   data required time
          -1.08   data arrival time
-----------------------------------------------------------
```

Appendix D — Full 18-corner terminal run output — continued (lines 1786 onward)

```
          1.35   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -39.70
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max -7993.79
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ss_n40C_1v35

============================================================
RUNNING CORNER: ss_n40C_1v40
============================================================

############################################################
CORNER: ss_n40C_1v40
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v40.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v40.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v40.lib line 24, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_n40C_1v40 ====="
===== CORNER: ss_n40C_1v40 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.94    0.94 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.43    1.38 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.34    1.72 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   1.70    3.42 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.99    4.40 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.87    5.27 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.44    5.71 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.71    6.43 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.44    6.87 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.71    7.58 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.44    8.02 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.71    8.74 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.44    9.18 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.71    9.89 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.44   10.33 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.71   11.05 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
```

```
0.44   11.49 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.71   12.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.44   12.64 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.71   13.36 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.44   13.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.66   14.46 v core/U96/X (sky130_fd_sc_hd__a21o_1)
1.95   16.41 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.74   17.14 v core/U94/X (sky130_fd_sc_hd__a21o_1)
1.91   19.05 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
1.98   21.04 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
```

Appendix D — Full 18-corner terminal run output — continued (lines 1891 onward)

```
   2.00   23.03 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   1.79   24.82 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.22   25.04 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.14   25.18 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   1.92   27.10 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   1.98   29.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   1.98   31.07 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   2.02   33.10 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.74   33.83 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   1.91   35.74 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   2.46   38.20 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   1.00   39.20 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.53   39.74 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00   39.74 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          39.74   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.96   10.04   library setup time
          10.04   data required time
-----------------------------------------------------------
          10.04   data required time
         -39.74   data arrival time
-----------------------------------------------------------
         -29.69   slack (VIOLATED)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -29.69
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max -5466.48
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ss_n40C_1v40

============================================================
RUNNING CORNER: ss_n40C_1v44
============================================================

############################################################
CORNER: ss_n40C_1v44
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v44.lib
############################################################
```

```
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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
```

Appendix D — Full 18-corner terminal run output — continued (lines 1996 onward)

```
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v44.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v44.lib line 24, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_n40C_1v44 ====="
===== CORNER: ss_n40C_1v44 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.83    0.83 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.38    1.21 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.30    1.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   1.50    3.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.88    3.88 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.77    4.65 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.39    5.04 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.62    5.67 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.39    6.06 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.62    6.68 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.39    7.07 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.62    7.69 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.39    8.08 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.62    8.71 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.39    9.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.62    9.72 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.39   10.11 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.62   10.73 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.39   11.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.62   11.75 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.39   12.14 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.57   12.71 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   1.65   14.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.63   15.00 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   1.61   16.61 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   1.67   18.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   1.68   19.96 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   1.51   21.48 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.19   21.67 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.13   21.79 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   1.62   23.42 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   1.67   25.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   1.67   26.76 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   1.71   28.47 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.63   29.10 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   1.61   30.72 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   2.07   32.79 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.87   33.65 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.45   34.10 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00   34.10 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          34.10   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.80   10.20   library setup time
          10.20   data required time
---------------------------------------------------------
          10.20   data required time
         -34.10   data arrival time
---------------------------------------------------------
         -23.91   slack (VIOLATED)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
```

```
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.80    0.80 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
```

Appendix D — Full 18-corner terminal run output — continued (lines 2101 onward)

```
   0.00    0.80 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.80   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.19   -0.19   library hold time
          -0.19   data required time
-----------------------------------------------------------
          -0.19   data required time
          -0.80   data arrival time
-----------------------------------------------------------
           0.99   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -23.91
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max -4107.19
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ss_n40C_1v44

============================================================
RUNNING CORNER: ss_n40C_1v60
============================================================

############################################################
CORNER: ss_n40C_1v60
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v60.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v60.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v60.lib line 24, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_n40C_1v60 ====="
===== CORNER: ss_n40C_1v60 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.57    0.57 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.25    0.82 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.20    1.02 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
```

```
1.04    2.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.53    2.58 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.52    3.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.24    3.35 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.41    3.76 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.24    4.00 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.41    4.41 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.24    4.66 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.41    5.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.24    5.31 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
```

Appendix D — Full 18-corner terminal run output — continued (lines 2206 onward)

```
   0.41    5.72 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    5.96 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    6.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    6.62 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    7.03 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    7.27 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    7.68 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.25    7.93 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    8.30 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.98    9.28 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.40    9.68 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.96   10.64 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.99   11.63 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   1.00   12.63 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.91   13.53 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.12   13.65 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.09   13.74 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.96   14.71 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.99   15.70 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.99   16.69 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   1.02   17.71 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.40   18.11 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.96   19.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   1.21   20.27 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.59   20.87 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.29   21.15 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00   21.15 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          21.15   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.46   10.54   library setup time
          10.54   data required time
-----------------------------------------------------------
          10.54   data required time
         -21.15   data arrival time
-----------------------------------------------------------
         -10.61   slack (VIOLATED)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.54    0.54 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.54 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.54   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.12   -0.12   library hold time
          -0.12   data required time
---------------------------------------------------------
          -0.12   data required time
          -0.54   data arrival time
---------------------------------------------------------
           0.66   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -10.61
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max -1355.80
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew
```

```
%
% exit
CORNER_STATUS: PASS - ss_n40C_1v60

============================================================
RUNNING CORNER: ss_n40C_1v60_ccsnoise
============================================================

############################################################
```

Appendix D — Full 18-corner terminal run output — continued (lines 2311 onward)

```
CORNER: ss_n40C_1v60_ccsnoise
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v60_ccsnoise.lib
############################################################

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This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<c_hd/lib/sky130_fd_sc_hd__ss_n40C_1v60_ccsnoise.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v60_ccsnoise.lib l
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_n40C_1v60_ccsnoise ====="
===== CORNER: ss_n40C_1v60_ccsnoise =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.57    0.57 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.25    0.82 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.20    1.02 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   1.04    2.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.53    2.58 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.52    3.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    3.35 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    3.76 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    4.00 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    4.41 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    4.66 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    5.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    5.31 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    5.72 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    5.96 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    6.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    6.62 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    7.03 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    7.27 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.41    7.68 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.25    7.93 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    8.30 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.98    9.28 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.40    9.68 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.96   10.64 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.99   11.63 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   1.00   12.63 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.91   13.53 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.12   13.65 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.09   13.74 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.96   14.71 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.99   15.70 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.99   16.69 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   1.02   17.71 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.40   18.11 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.96   19.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   1.21   20.27 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.59   20.87 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.29   21.15 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00   21.15 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          21.15   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.46   10.54   library setup time
          10.54   data required time
---------------------------------------------------------
```

```
          10.54   data required time
         -21.15   data arrival time
-------------------------------------------------------
         -10.61   slack (VIOLATED)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
```

Appendix D — Full 18-corner terminal run output — continued (lines 2416 onward)

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.54    0.54 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.54 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.54   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.12   -0.12   library hold time
          -0.12   data required time
---------------------------------------------------------
          -0.12   data required time
          -0.54   data arrival time
---------------------------------------------------------
           0.66   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -10.61
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max -1355.80
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ss_n40C_1v60_ccsnoise

============================================================
RUNNING CORNER: ss_n40C_1v76
============================================================

############################################################
CORNER: ss_n40C_1v76
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v76.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v76.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__ss_n40C_1v76.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: ss_n40C_1v76 ====="
===== CORNER: ss_n40C_1v76 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
```

```
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
```

Appendix D — Full 18-corner terminal run output — continued (lines 2521 onward)

```
   0.43    0.43 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.19    0.62 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.15    0.77 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.80    1.57 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.36    1.93 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.39    2.32 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    2.50 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.31    2.80 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    2.98 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.31    3.29 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    3.47 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.31    3.77 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    3.95 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.31    4.26 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    4.44 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.31    4.74 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    4.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.31    5.23 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    5.41 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.31    5.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    5.89 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.27    6.16 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.68    6.84 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.29    7.13 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.65    7.79 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.68    8.47 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.68    9.15 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.63    9.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.09    9.87 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.07    9.94 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.66   10.61 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.68   11.28 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.68   11.96 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.70   12.66 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.29   12.95 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.65   13.61 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.82   14.43 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.45   14.88 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.19   15.07 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00   15.07 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          15.07   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.31   10.69   library setup time
          10.69   data required time
-----------------------------------------------------------
          10.69   data required time
         -15.07   data arrival time
-----------------------------------------------------------
          -4.38   slack (VIOLATED)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.41    0.41 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.41 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.41   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.09   -0.09   library hold time
          -0.09   data required time
-----------------------------------------------------------
          -0.09   data required time
          -0.41   data arrival time
-----------------------------------------------------------
           0.50   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max -4.38
%
% puts "===== TNS ====="
```

```
===== TNS =====
% report_tns
tns max -367.93
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
```

Appendix D — Full 18-corner terminal run output — continued (lines 2626 onward)

```
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - ss_n40C_1v76

============================================================
RUNNING CORNER: tt_025C_1v80
============================================================

############################################################
CORNER: tt_025C_1v80
LIBRARY: sky130_fd_sc_hd__tt_025C_1v80.lib
############################################################

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This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__tt_025C_1v80.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__tt_025C_1v80.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: tt_025C_1v80 ====="
===== CORNER: tt_025C_1v80 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.19    3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.39    4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.18    4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.37    4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.38    5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.36    5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.37    6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.38    6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.38    7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.40    7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.18    7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.37    8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
```

```
  0.45    8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
  0.29    8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
  0.10    8.90 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
  0.00    8.90 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
          8.90   data arrival time

 11.00   11.00   clock clk (rise edge)
  0.00   11.00   clock network delay (ideal)
  0.00   11.00   clock reconvergence pessimism
         11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
```

Appendix D — Full 18-corner terminal run output — continued (lines 2731 onward)

```
  -0.16   10.84   library setup time
          10.84   data required time
-----------------------------------------------------------
          10.84   data required time
          -8.90   data arrival time
-----------------------------------------------------------
           1.94   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.27    0.27 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.27 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.27   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.03   -0.03   library hold time
          -0.03   data required time
---------------------------------------------------------
          -0.03   data required time
          -0.27   data arrival time
---------------------------------------------------------
           0.31   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - tt_025C_1v80

============================================================
RUNNING CORNER: tt_100C_1v80
============================================================

############################################################
CORNER: tt_100C_1v80
LIBRARY: sky130_fd_sc_hd__tt_100C_1v80.lib
############################################################

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under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
% set_units -time ns
Warning 345: time scale 1e-09 does not match library scale 1e+00.
%
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
1
% read_liberty /data/Desktop/VSDBabySoC/src/lib/avsddac.lib
1
<y130_fd_sc_hd/lib/sky130_fd_sc_hd__tt_100C_1v80.lib
Warning 1171: /data/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__tt_100C_1v80.lib line 23, d
1
%
% read_verilog /data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
1
```

```
% link_design vsdbabysoc
1
%
% read_sdc /data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
%
% puts "===== CORNER: tt_100C_1v80 ====="
===== CORNER: tt_100C_1v80 =====
%
% puts "===== SETUP (MAX) ====="
===== SETUP (MAX) =====
```

Appendix D — Full 18-corner terminal run output — continued (lines 2836 onward)

```
<max -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.29    0.29 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
   0.11    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
   0.10    0.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
   0.48    0.99 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
   0.19    1.18 v core/U479/X (sky130_fd_sc_hd__xor2_1)
   0.23    1.41 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
   0.12    1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    1.71 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    1.83 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    2.01 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.12 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    2.31 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.42 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    2.61 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    2.72 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    2.90 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.02 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    3.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.32 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
   0.18    3.50 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
   0.11    3.62 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
   0.17    3.79 v core/U96/X (sky130_fd_sc_hd__a21o_1)
   0.38    4.16 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
   0.17    4.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)
   0.36    4.69 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
   0.37    5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
   0.37    5.43 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
   0.35    5.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
   0.05    5.83 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
   0.06    5.89 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
   0.36    6.26 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
   0.37    6.62 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
   0.37    6.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
   0.39    7.38 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
   0.17    7.55 v core/U81/X (sky130_fd_sc_hd__a21o_1)
   0.36    7.91 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
   0.49    8.40 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
   0.25    8.65 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
   0.16    8.81 ^ core/U684/Y (sky130_fd_sc_hd__o22ai_1)
   0.00    8.81 ^ core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
           8.81   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.00   11.00   clock network delay (ideal)
   0.00   11.00   clock reconvergence pessimism
          11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.10   10.90   library setup time
          10.90   data required time
---------------------------------------------------------
          10.90   data required time
          -8.81   data arrival time
---------------------------------------------------------
           2.09   slack (MET)


%
% puts "===== HOLD (MIN) ====="
===== HOLD (MIN) =====
<min -group_path_count 1 -format full_clock_expanded
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.28 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
           0.28   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.04   -0.04   library hold time
          -0.04   data required time
---------------------------------------------------------
```

```
          -0.04   data required time
          -0.28   data arrival time
-------------------------------------------------------
           0.31   slack (MET)


%
% puts "===== WNS ====="
===== WNS =====
% report_wns
```

Appendix D — Full 18-corner terminal run output — continued (lines 2941 onward)

```
wns max 0.00
%
% puts "===== TNS ====="
===== TNS =====
% report_tns
tns max 0.00
%
% puts "===== CLOCK SKEW ====="
===== CLOCK SKEW =====
% report_clock_skew
Clock clk
   0.00 source latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 target latency core/CPU_Xreg_value_a4_reg[10][0]/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

%
% exit
CORNER_STATUS: PASS - tt_100C_1v80

============================================================
ALL 18 CORNERS COMPLETED
RESULT FILE:
/home/vsduser/Desktop/VSDBabySoC/18_corner_results.txt
============================================================
vsduser@vsdsquadron:~/Desktop/VSDBabySoC$
```

10. Reproducibility Notes

The terminal evidence shows the project was run from ~/Desktop/VSDBabySoC inside the Ubuntu VM. OpenSTA was launched through Docker with the home directory mounted to /data. The multi-corner script generated a result file under the VSDBabySoC directory and printed the final “ALL 18 CORNERS COMPLETED” message.

```
docker run --rm -it -v "$HOME":/data opensta

/data/Desktop/VSDBabySoC/src/lib/vsdbabysoc.synth.v
/data/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
/data/Desktop/VSDBabySoC/src/lib/avsdpll.lib
/data/Desktop/VSDBabySoC/src/lib/avsddac.lib
```

OpenSTA command names used in the final run were report_checks, report_wns, report_tns and report_clock_skew. The earlier terminal evidence also shows the initial network-not-linked condition before the standard-cell timing library was loaded and the design was linked successfully.

11. Conclusion

The VSDBabySoC synthesized design was analyzed using an 11 ns clock over the recorded 18 SKY130 timing-library cases. The final run completed all 18 cases. The main recorded issue is setup timing under SS conditions, while the recorded hold slacks remain positive. The report preserves both the successful results and the debugging steps so that the analysis can be followed from the terminal evidence through the final numerical table.

End of report — B.Rohith Reddy (25EG504B06)